

Thoughts on Large Foundation Models

Patrick H. Chen

October 23, 2025

Lecture 16

Emergence



<https://www.newscientist.com/article/mg25834382-300-emergence-the-mysterious-concept-that-holds-the-key-to-consciousness/>

Deepseek V3: 63M $\Rightarrow$ 6M

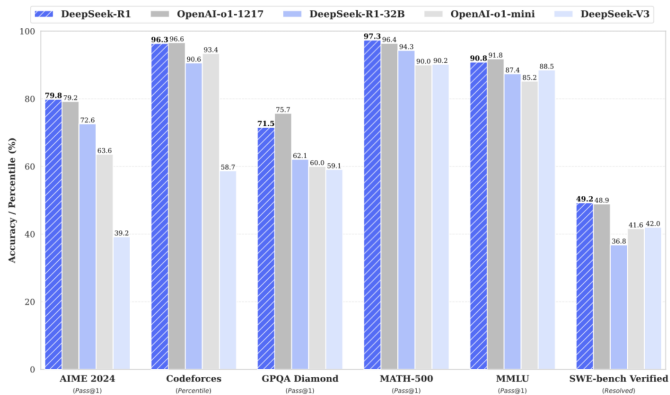


Figure 1 | Benchmark performance of DeepSeek-R1.

GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models

GSM8K

When Sophie watches her nephew, she gets out a variety of toys for him. The bag of building blocks has 31 blocks in it. The bin of stuffed animals has 8 stuffed animals inside. The tower of stacking rings has 9 multicolored rings on it. Sophie recently bought a tube of bouncy balls, bringing her total number of toys for her nephew up to 62. How many bouncy balls came in the tube?

Let T be the number of bouncy balls in the tube.
After buying the tube of balls, Sophie has $31+8+9+T = 48+T=62$ toys for her nephew.
Thus, $T = 62-48 = \langle\langle 62-48-14 \rangle\rangle 14$ bouncy balls came in the tube.

GSM Symbolic Template

When {name} watches her {family}, she gets out a variety of toys for him. The bag of building blocks has {x} blocks in it. The bin of stuffed animals has {y} stuffed animals inside. The tower of stacking rings has {z} multicolored rings on it. {name} recently bought a tube of bouncy balls, bringing her total number of toys she bought for her {family} up to {total}. How many bouncy balls came in the tube?

#variables:

```
- name = sample(names)
- family = sample(["nephew", "cousin", "brother"])
- x = range(5, 100)
- y = range(5, 100)
- z = range(5, 100)
- total = range(100, 500)
- ans = range(85, 200)
```

#conditions:

```
- x + y + z + ans == total
```

Let T be the number of bouncy balls in the tube. After buying the tube of balls, {name} has {x} + {y} + {z} + T = {x + y + z} + T = {total} toys for her {family}.

Thus, $T = \{total\} - \{x + y + z\} = \langle\langle \{total\} - \{x + y + z\} - \{ans\} \rangle\rangle \{ans\}$ bouncy balls came in the tube.

Figure 1: Illustration of the GSM-Symbolic template creation process. This dataset serves as a tool to investigate the presumed reasoning capabilities of LLMs, enabling the design of controllable mathematical reasoning evaluations with more reliable metrics. Our results reveal that all state-of-the-art LLMs exhibit significant performance variations, suggesting the fragility or lack of reasoning.

GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models

GSM-NoOp

Oliver picks 44 kiwis on Friday. Then he picks 58 kiwis on Saturday. On Sunday, he picks double the number of kiwis he did on Friday, **but five of them were a bit smaller than average**. How many kiwis does Oliver have?

o1-mini: Oliver picks kiwis over three days as follows:

- Friday: 44 kiwis

- Saturday: 58 kiwis

- Sunday: Double the number he picked on Friday, which is $2 \times 44 = 88$ kiwis

However, on Sunday, 5 of these kiwis were smaller than average. We need to subtract them from the Sunday total: 88 (Sunday's kiwis) - 5 (smaller kiwis) = 83 kiwis Now, summing up the kiwis from all three days: 44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis Oliver has a total of 185 kiwis.

Llama3-8B: Let's break it down step by step:

Oliver picks 44 kiwis on Friday. He picks 58 kiwis on Saturday. On Sunday, he picks double the number of kiwis he did on Friday, which is $2 \times 44 = 88$ kiwis.

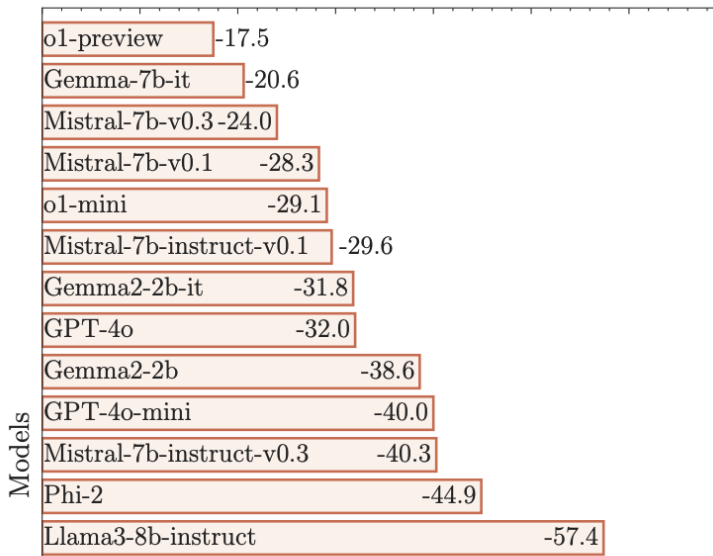
Five of the kiwis picked on Sunday are a bit smaller than average, so we subtract 5 from the total number of kiwis picked on Sunday: $88 - 5 = 83$ kiwis. Now, let's add up the total number of kiwis Oliver has:

44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis

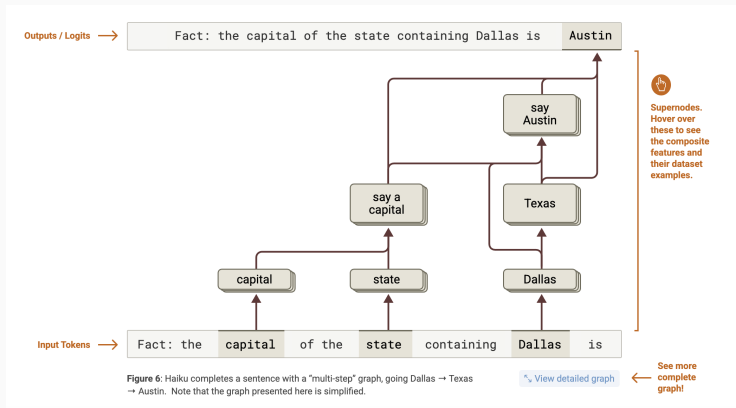
So, Oliver has 185 kiwis in total.

Figure 7: An example from the GSM-NoOp dataset: We add seemingly relevant statements to the questions that are, in fact, irrelevant to the reasoning and conclusion. However, the majority of models fail to ignore these statements and blindly convert them into operations, leading to mistakes.

GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models



Claude Model Explanation



<https://transformer-circuits.pub/2025/attribution-graphs/biology.html>

Claude Model Explanation

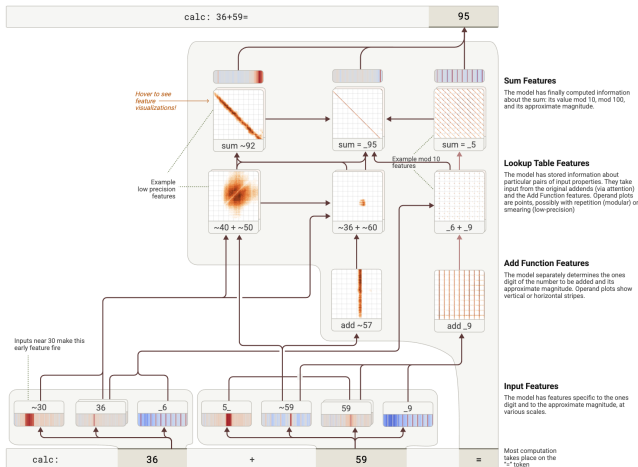


Figure 26. A simplified attribution graph of Haiku adding two-digit numbers. Features of the inputs feed into separable processing pathways.

Claude Model Explanation

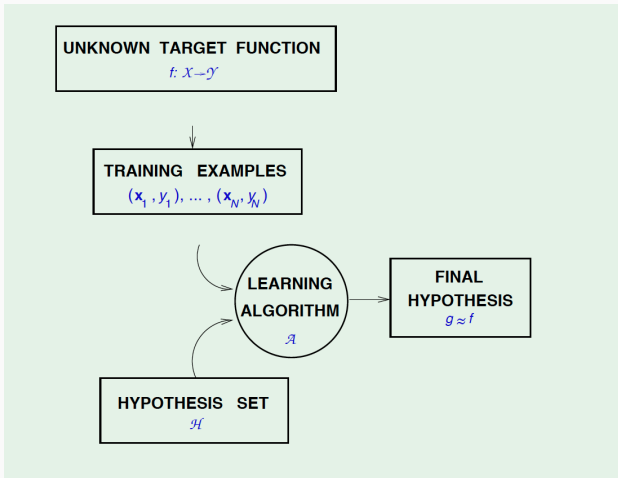
Human: Answer in one word. What is $36+59$?

Assistant: 95

Human: Briefly, how did you get that?

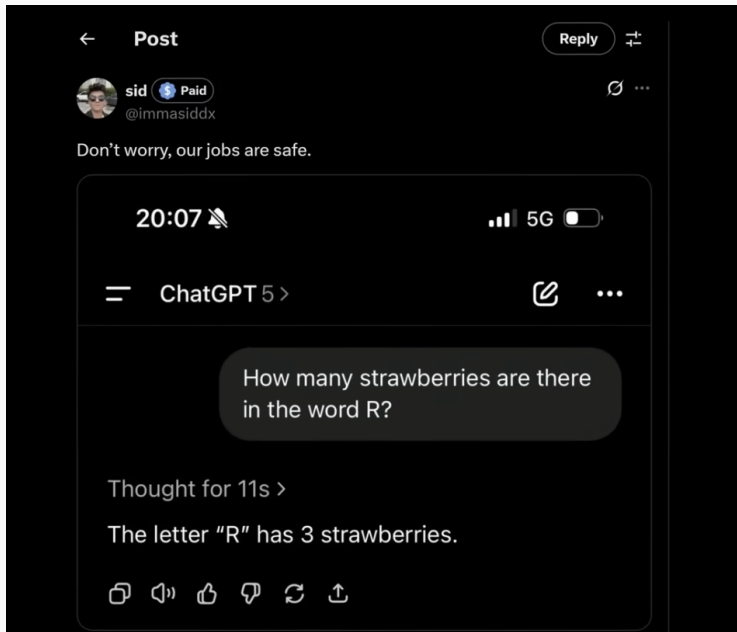
Assistant: I added the ones ($6+9=15$), carried the 1, then added the tens ($3+5+1=9$), resulting in 95.

OOD Problem is still there.



(Figure from “Learning from Data”)

Can't make sense of Nonsense



Security Issues

